SSH-Net: Symplectic-Inspired Discrete Dynamic Evolution for Hyperspectral Image Classification

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***Abstract*—Hyperspectral image classification (HSIC) relies on effective modeling of coupled spectral–spatial interactions. While recent CNN-, Transformer-, and Mamba-based methods have improved feature extraction, most of them organize token interaction through layerwise aggregation and repeated stacking, leaving cross-layer propagation only implicitly modeled. To address this issue, we propose SSH-Net, a symplectic-inspired spectral–spatial dynamic interaction network for HSIC. The proposed framework parameterizes token representations as latent position and momentum states and performs gated leapfrog-style dynamic updates to realize structured multistep feature propagation. On top of this dynamic backbone, a terminal-state descriptor branch summarizes the final position, momentum, and interaction-response statistics as terminal-state evidence, and a discriminative evidence aggregation (DEA) branch integrates such evidence with multiscale spatial and global contextual features for prediction. In this way, SSH-Net combines Hamiltonian-inspired dynamic propagation with complementary state-aware readout for hyperspectral classification. Experiments on four public benchmarks, namely Indian Pines, Houston2013, WHU-Hi-LongKou, and WHU-Hi-HanChuan, show that SSH-Net achieves consistently competitive or superior performance in terms of overall accuracy (OA), average accuracy (AA), and kappa coefficient compared with representative baseline methods. These results suggest that symplectic-inspired dynamic propagation provides a useful inductive bias for organizing spectral–spatial interaction in HSIC. Codes are available at https://github.com/yi1275174812/SSH-Net**

***Index Terms*—Deep learning, hyperspectral image classification (HSIC), symplectic-inspired dynamic propagation, Transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) jointly encode spatial and spectral information, where each pixel is associated with a contiguous spectral signature that enables fine-grained material characterization. Owing to their rich spectral content, HSIs have been widely used in precision agriculture [1], mineral exploration [2], water quality monitoring [3], and urban land-cover mapping [4]. HSI classification (HSIC) exploits such spectral–spatial information to assign pixels to semantic categories and remains a core task in remote sensing. However, the intrinsic spectral–spatial complexity of HSIs makes deep cross-layer updating difficult to organize coherently, leading to inconsistent representation accumulation and weakened discriminative structure.

Deep learning has become a mainstream paradigm for HSIC and has substantially advanced spectral–spatial representation learning [5], [6]. Existing HSIC models have mainly evolved along three directions. CNN-based methods exploit local connectivity for neighborhood-aware spectral–spatial modeling [7], [8], [9]. Representative methods such as 3D-CNN and HybridSN learn joint spectral–spatial features directly from hyperspectral cubes or hybrid convolutional pipelines, providing effective local representation learning [10], [11]. However, CNN-based methods still extend interactions mainly through local aggregation and receptive-field stacking, so long-range dependency modeling remains indirect. Transformer-based methods introduce content-adaptive token interaction and improve nonlocal dependency modeling [12], [13], [14]. Representative models such as SpectralFormer, spectral–spatial feature tokenization Transformer (SSFTT), MorphFormer, and DSFormer further enhance spectral–spatial interaction through spectral tokenization, tailored token construction, morphological modeling, and dual-branch fusion [13], [15], [16], [17]. Yet most Transformer-based methods still rely on one-step similarity-weighted aggregation, with deeper interaction formed implicitly by repeated stacking. More recently, Mamba-based methods have provided an efficient alternative through selective state-space scanning [18], [19], [20]. Representative methods such as SS-Mamba, MFormer, and MambaHSI perform spectral–spatial modeling through stacked state-space blocks, hybrid Mamba–Transformer interaction, or decoupled spatial–spectral state modeling [21], [22], [23]. However, current Mamba-based methods still depend heavily on tokenization, scan order, and adaptation design, leaving propagation insufficiently explicit and controllable.

From a broader perspective, dynamical-system formulations treat deep feature propagation as structured state evolution, implying that representation quality depends not only on model capacity but also on the propagation law itself [24],

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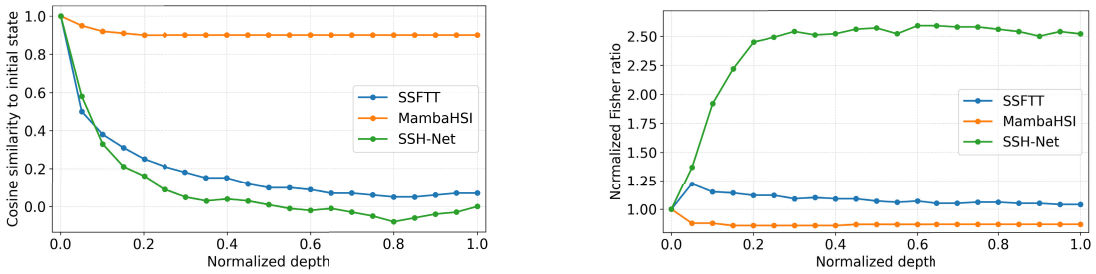


Fig. 1. Layerwise representation evolution across different backbones. SSH-Net couples stronger cross-layer evolution with sustained discriminability improvement, while SSFTT and MambaHSI show more limited refinement.

[25], [26], [27]. This view is particularly relevant to HSIC, where spectral–spatial interaction must remain coherent and discriminative across layers, yet it has not been developed into an explicit paradigm for token-level interaction. As shown in Fig. 1, SSFTT (Transformer-based) shows substantial deviation from the initial state but only limited discriminability improvement, while MambaHSI (Mamba-based) remains overly conservative with similarly weak refinement. On the contrary, SSH-Net achieves more sufficient cross-layer transition together with sustained discriminative enhancement, suggesting that hyperspectral interaction modeling can benefit from more explicitly organized propagation to avoid both ineffective drift and overly conservative updates.

Synthesizing the above progress, current HSIC methods still face three interaction modeling limitations.

1) Most methods rely on static layerwise transformation or one-step aggregation, so cross-layer dependency is formed implicitly, and multistep feature evolution remains weakly modeled.
2) Deep propagation lacks explicit control over update magnitude, interaction strength, and propagation scale, making feature evolution difficult to keep consistent and stable in challenging hyperspectral scenes.
3) Existing state-space and dynamical methods have not yet formed a dedicated interaction paradigm for hyperspectral representation learning, where token-level spectral–spatial coupling is organized through an explicit dynamic update form and directly linked to classification evidence.

Motivated by the above observations, we investigate whether a symplectic-inspired dynamic formulation can provide a more structured way to organize spectral–spatial propagation in HSIC. The inherent suitability lies in two critical insights. First, the coupled position–momentum decomposition naturally separates the spectral–spatial representation profiles from their cross-layer directional updating tendencies. Second, the volume-preserving geometric intuition associated with symplectic-inspired structures provides a useful prior for regulating feature propagation, which may help alleviate feature over-smoothing and representation drift in deep architectures while preserving fine-grained discriminative cues. To this end, we propose SSH-Net, a symplectic spectral–spatial Hamiltonian network. Instead of treating token interaction as one-shot similarity aggregation, SSH-Net models token representations as latent position and momentum states and updates them through gated leapfrog-style transitions, thereby enabling structured multistep propagation across layers. Built upon this dynamic backbone, the proposed framework further introduces a symplectic state descriptor (SSD) branch to summarize the final position, momentum, and interaction-response statistics as compact terminal-state evidence, and a discriminative evidence aggregation (DEA) branch to combine such evidence with multiscale spatial and global contextual cues for prediction. In this way, SSH-Net integrates structured dynamic propagation with state-aware feature readout for hyperspectral classification.

The main contributions of this article are summarized as follows.

1) We propose SSH-Net, a symplectic-inspired dynamic interaction network for HSIC, which models spectral–spatial token propagation with coupled position and momentum states. To the best of our knowledge, this work presents an early attempt to introduce a symplectic-inspired dynamic formulation into HSIC.
2) We develop a symplectic Hamiltonian dynamics (SHDs) block to perform Hamiltonian-inspired position–momentum decomposition and gated leapfrog-style updates, enabling structured multistep spectral–spatial interaction across layers.
3) We introduce an SSD and a DEA module to summarize terminal latent-state statistics and fuse them with contextual features for classification.
4) Experiments on four public hyperspectral datasets show that SSH-Net achieves consistently competitive or superior classification performance, while ablation studies support the effectiveness of the proposed design.

II. Related Work

A. Mainstream Architecture-Based Methods for HSIC

1) CNN-Based Methods for HSIC: CNN-based methods remain a major paradigm for HSIC because convolution naturally exploits local neighborhood structure, parameter sharing, and hierarchical feature abstraction for spectral–spatial representation learning [11]. Hu et al. [28] introduced deep CNNs into HSIC by operating directly in the spectral domain, showing that hierarchical convolutional processing can already provide effective spectral discrimination. Li et al. [10] subsequently proposed a 3D-CNN framework that performs convolution directly over hyperspectral cubes, enabling local

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spatial structure and adjacent spectral responses to be encoded within a unified operator. Yang et al. [29] proposed SyCNN to coordinate 2-D and 3-D convolutions and to strengthen spectral–spatial data interaction within a hybrid convolutional framework. Jin et al. [30] proposed a dual heterogeneous network with spatial-specific and spectral-specific branches, spectral–spatial involution, and feature calibration to capture complementary spectral–spatial information and reduce neighboring-pixel interference. Xi et al. [31] further proposed DGSSC, which couples a 3-D/2-D convolutional encoder with a deep latent variable model to enhance spectral–spatial representation under imbalanced and limited-sample settings.

Overall, CNN-based HSIC methods provide strong local spectral–spatial representation learning through convolutional coupling and multiscale feature extraction. However, their interaction mechanism is inherently constrained by local receptive fields, making long-range dependency modeling rely on deeper stacking or auxiliary modules rather than direct interaction.

2) Transformer-Based Methods for HSIC: Transformer-based methods provide a more direct mechanism for HSIC by modeling spectral–spatial dependencies through content-adaptive token interaction [32], [33]. Hong et al. [15] developed SpectralFormer and reformulated HSIC from a spectral-sequence perspective, using groupwise spectral embedding together with cross-layer skip fusion to strengthen spectral dependency modeling and information transmission. Sun et al. [13] subsequently proposed SSFTT, where shallow spectral–spatial features are first extracted by convolutional layers and then reorganized by a Gaussian-weighted tokenizer for Transformer encoding, making token construction itself part of hyperspectral representation design. Roy et al. [16] further proposed MorphFormer, which introduces learnable spectral and spatial morphological operations into the Transformer framework to enhance the interaction of structural and shape information in hyperspectral tokens. Yang et al. [34] proposed the Quaternion Transformer Network (QTN), which combines band-adaptive spectral selection with quaternion attention to capture spectral–spatial correlations and long-range dependencies more effectively. More recently, Xu et al. [17] introduced DSFormer, which performs joint spatial–spectral contextual modeling by adaptively selecting and fusing features across different receptive fields and further suppresses irrelevant information through token-selective fusion.

In summary, Transformer-based HSIC methods substantially improve nonlocal dependency modeling and token interaction flexibility through content-adaptive attention. Nevertheless, spectral–spatial interaction is still realized as one-step similarity-weighted aggregation within each layer, while deeper interaction is formed implicitly through stacking rather than being explicitly organized across layers.

3) Mamba-Based Methods for HSIC: Mamba-based methods have recently become an important direction in HSIC because selective state-space modeling provides linear-complexity sequence modeling while retaining the ability to capture long-range spectral–spatial dependency [21], [35]. Huang et al. [21] proposed SS-Mamba and introduced spectral and spatial token generation followed by stacked spectral–spatial Mamba blocks, providing an early Mamba-based framework for joint hyperspectral feature modeling. Yao et al. [35] subsequently proposed SpectralMamba, which combines dynamical masking and piecewise spectral scanning to improve efficient sequence modeling under high spectral dimensionality. He et al. [22] further proposed MFormer, which integrates Mamba and Transformer experts within a hybrid multiscale framework, using multiscale token generation, inner-scale Transformer modeling for short-range dependencies, and cross-scale bidirectional Mamba modeling for long-range interactions across token groups. Li et al. [23] also proposed MambaHSI, which decouples spatial and spectral modeling at the image level and then performs adaptive fusion to improve global spectral–spatial representation.

Collectively, Mamba-based HSIC methods improve the efficiency of long-range spectral–spatial modeling and expand the backbone choices for hyperspectral representation learning. However, they primarily model interaction through sequential state propagation, where dependency is shaped by scan order and state transition design, making token interaction less directly controllable compared with explicit pairwise modeling.

4) Attention-Augmented Methods for HSIC: In addition to the above backbone-driven paradigms, a line of work enhances spectral–spatial modeling by integrating attention mechanisms into convolutional or hybrid architectures. AMGCFN [36] combines multiscale convolutional features with graph-based attention to capture long-range dependencies and improve structural representation. MSNAT [37] introduces multiscale neighborhood attention to strengthen local interaction modeling across different spatial ranges. CACFTNet [38] further incorporates covariance-based attention together with cross-layer fusion to enhance hierarchical spectral–spatial representation.

These methods enhance spectral–spatial representation by introducing attention into convolutional or hybrid frameworks. However, they mainly act as auxiliary enhancement mechanisms on top of existing architectures, without fundamentally changing how interaction is organized, which remains dominated by layerwise aggregation.

Recent studies in hyperspectral and general remote sensing tasks have explored prior-guided and structure-aware representation learning in several forms. In hyperspectral learning, multiteacher multiobjective meta-learning has been used for zero-shot heterogeneous band selection [39], while shift-sensitive spatial–spectral disentangling has been developed for unsupervised domain adaptation [40]. In remote sensing image super resolution, conditional diffusion-assisted hybrid attention [41] and semantic-guided generative diffusion priors [42] have also been investigated. These studies indicate that prior-guided and structure-aware representation modeling has become an important direction for organizing spectral–spatial and structural information in remote sensing tasks.

B. Dynamical-System-Based Methods for HSIC

In broader deep learning research, dynamical-system formulations have provided an alternative view of deep feature

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propagation by interpreting layerwise updating as state evolution over depth. Chen et al. [24] proposed neural ordinary differential equations (neural ODEs) and recast deep networks as continuous-depth systems governed by ordinary differential equations. Greydanus et al. [25] further proposed Hamiltonian neural networks and showed that the Hamiltonian formulation can impose more explicit structural priors on neural modeling. Zhong et al. [26] developed symplectic ODE-Net and demonstrated that symplectic integration helps preserve geometric structure during long-horizon propagation. These studies suggest that deep representation learning is shaped not only by network architecture but also by the propagation law used to organize feature evolution. In HSIC, related exploration remains very limited. Paoletti et al. [43] made an early and meaningful attempt to introduce neural ODEs into HSIC by interpreting residual updating as continuous-depth evolution.

These studies indicate that feature propagation can be viewed not only as an architectural effect, but also as a dynamical process governed by an explicit update rule. In HSIC, however, such exploration remains limited, and existing methods still lack a dynamic interaction framework that explicitly organizes token-level spectral–spatial propagation across layers.

To address this gap, we propose SSH-Net, which models hyperspectral token interaction as symplectic-inspired dynamic propagation with coupled position–momentum states and links the evolved terminal states to classification through descriptor and aggregation modules. To the best of our knowledge, SSH-Net is an early symplectic-inspired dynamic interaction framework for HSIC.

III. Preliminaries of Hamiltonian Dynamics and Symplectic Integration

This section introduces the Hamiltonian and symplectic-integration formulations that motivate the proposed dynamic interaction design. From this viewpoint, deep spectral–spatial propagation can be interpreted as a structured multistep state evolution process, in which coupled position and momentum variables provide an explicit state description and symplectic discretization offers a principled update template. These preliminaries are introduced to establish the mathematical foundation for the proposed propagation strategy, rather than to claim that the full network constitutes a strictly physical Hamiltonian system.

Let the system state be

$$\mathbf{z}(t) = (\mathbf{q}(t), \mathbf{p}(t)) \tag{1}$$

where $\mathbf{q}(t)$ and $\mathbf{p}(t)$ denote the position and momentum states, respectively. The system is governed by the separable Hamiltonian

$$\mathcal{H}(\mathbf{q}, \mathbf{p}) = \mathcal{T}(\mathbf{p}) + \mathcal{V}(\mathbf{q}) \tag{2}$$

where $\mathcal{T}(\mathbf{p})$ and $\mathcal{V}(\mathbf{q})$ are the kinetic and potential terms. The corresponding canonical equations are

$$\dot{\mathbf{q}} = \frac{\partial \mathcal{H}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}}, \qquad \dot{\mathbf{p}} = -\frac{\partial \mathcal{H}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{q}}. \tag{3}$$

Introducing the canonical symplectic matrix

$$\mathbf{J} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{bmatrix} \tag{4}$$

the dynamics can be written compactly as

$$\dot{\mathbf{z}} = \mathbf{J}\nabla_{\mathbf{z}}\mathcal{H}(\mathbf{z}). \tag{5}$$

This form makes explicit that Hamiltonian evolution is a structured phase-space flow rather than an unconstrained vector-field update.

With the quadratic kinetic form

$$\mathcal{T}(\mathbf{p}) = \frac{1}{2}\|\mathbf{p}\|_2^2. \tag{6}$$

Equation (3) reduces to

$$\dot{\mathbf{q}} = \mathbf{p}, \qquad \dot{\mathbf{p}} = -\nabla_{\mathbf{q}}\mathcal{V}(\mathbf{q}). \tag{7}$$

Accordingly, $\mathbf{q}$ represents the current system configuration, whereas $\mathbf{p}$ characterizes its instantaneous update tendency and propagation direction. In the context of deep representation learning, this decomposition provides an interpretable dynamic view in which feature representations and their update trends are modeled jointly, and interaction-induced responses naturally act as the driving term of the coupled evolution.

A deep stacked architecture can be viewed as a discrete multistep feature evolution process, where stacking L propagation blocks yields a sequence of transitions from $(\mathbf{q}_0, \mathbf{p}_0)$ to $(\mathbf{q}_L, \mathbf{p}_L)$. The key design question is how to borrow a stable update form from Hamiltonian dynamics while keeping the model compatible with task-driven neural learning. In standard Hamiltonian systems, symplectic integration preserves the symplectic structure during discrete propagation. In SSH-Net, this property motivates the leapfrog-style update form used to organize latent feature propagation.

Symplectic integration is designed to preserve the symplectic geometry of Hamiltonian systems during discrete propagation. If $\Phi_{\Delta t}$ is symplectic, its Jacobian satisfies

$$\Phi'_{\Delta t}(\mathbf{z})^{\top}\mathbf{J}\Phi'_{\Delta t}(\mathbf{z}) = \mathbf{J}. \tag{8}$$

For standard Hamiltonian systems, this structure-preserving property makes the discrete trajectory more faithful to the qualitative behavior of the continuous flow over many steps. In this article, we use this property as the motivation for a structured leapfrog-style neural update, not as a proof of strict conservation in the complete classifier.

For the separable Hamiltonian in (2), this article adopts leapfrog as the practical symplectic integrator. Its update can be viewed as the symmetric splitting

$$\Phi_{\Delta t}^{\mathrm{LF}} = \Phi_{\mathcal{V}}^{\frac{\Delta t}{2}} \circ \Phi_{\mathcal{T}}^{\Delta t} \circ \Phi_{\mathcal{V}}^{\frac{\Delta t}{2}} \tag{9}$$

which provides a structure-preserving discrete realization of Hamiltonian evolution in the standard setting. Let the driving term induced by the potential field be

$$\mathbf{F}(\mathbf{q}) = \nabla_{\mathbf{q}}\mathcal{V}(\mathbf{q}) \tag{10}$$

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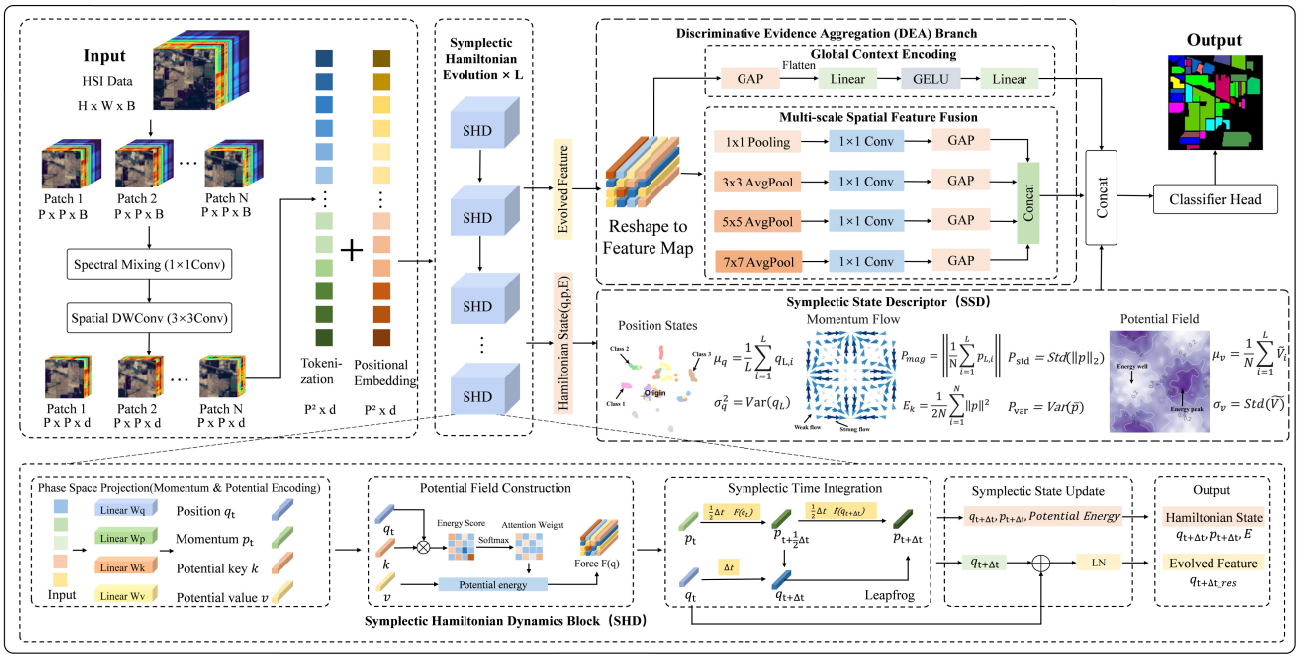


Fig. 2. Overall architecture of SSH-Net. The upper part illustrates the end-to-end information path from HSI patches to prediction. The lower part details the SHD block, including phase-space projection, potential-field construction, symplectic leapfrog time integration, state update, and the state-descriptor branch.

then the leapfrog propagation is

$$\mathbf{p}_{t+\frac{\Delta t}{2}} = \mathbf{p}_t - \frac{\Delta t}{2}\mathbf{F}(\mathbf{q}_t) \tag{11}$$

$$\mathbf{q}_{t+\Delta t} = \mathbf{q}_t + \Delta t\, \mathbf{p}_{t+\frac{\Delta t}{2}} \tag{12}$$

$$\mathbf{p}_{t+\Delta t} = \mathbf{p}_{t+\frac{\Delta t}{2}} - \frac{\Delta t}{2}\mathbf{F}\left(\mathbf{q}_{t+\Delta t}\right). \tag{13}$$

This update alternates momentum and position propagation within each step and evaluates the driving term at both $\mathbf{q}_t$ and $\mathbf{q}_{t+\Delta t}$, thereby providing a symmetric discrete update form that inspires the SHD implementation.

The Hamiltonian formulation and leapfrog discretization introduced above provide the inspiration and update template for organizing deep feature propagation as a coupled evolution process. In Section IV, this formulation is instantiated in the proposed network by defining an interaction-induced driving response, embedding the leapfrog-style update into the propagation module, and summarizing the resulting terminal latent states for classification.

IV. Proposed Approach

A. Overall Framework

This article presents SSH-Net, an end-to-end model for HSIC, as shown in Fig. 2. The framework consists of three components: the SHDs block, which serves as the evolution backbone for spectral–spatial interaction; the SSD branch, which reads out compact dynamical evidence from the terminal phase-space state; and the DEA branch, which complements the final-layer representation with multiscale spatial statistics and global contextual features. After L layers of SHD evolution, the final feature sequence is delivered to DEA, while the terminal position–momentum state and interaction response are summarized by SSD, and the resulting evidence is jointly used for prediction.

B. SHDs Block

1) Phase-Space Formulation and Discrete Evolution of SHD: Inspired by the Hamiltonian formulation and leapfrog-style discretization introduced in Section III, this section instantiates them in SSH-Net. The role of SHD is to provide an explicitly organized update mechanism for spectral–spatial propagation. By representing token features with latent position and momentum states, SHD makes the propagation process depend on an explicit state transition rule rather than only on repeated feature transformation. This formulation gives each layer a structured latent state and allows cross-layer interaction to be analyzed through the evolution of position, momentum, and interaction-response variables.

Given the input token sequence at layer t, $\mathbf{X}_t$, SHD applies linear projections to obtain the latent position state $\mathbf{q}_t$, latent momentum state $\mathbf{p}_t$, potential-field key $\mathbf{k}_t$, and potential-field value $\mathbf{v}_t$. Under the multihead implementation, the head dimension is set to $d_k = d/h$. SHD then computes the token interaction weights $\mathbf{A}(\mathbf{q}_t)$ from the scaled dot-product similarity between the position state and the potential-field key, and aggregates $\mathbf{v}_t$ to produce the potential-field response $\mathbf{V}(\mathbf{q}_t)$. This response is a tensorized interaction-induced driving field generated by attention-style similarity, rather than a scalar physical potential. When $d_v \neq d_k$, it is first aligned by a linear mapping and then used as the driving term $\mathbf{F}(\mathbf{q}_t)$.

Based on this driving term, SHD adopts the leapfrog scheme in (11)–(13) as a leapfrog-style update template and further introduces a learnable step size Δt and a learnable gating coefficient g to modulate the propagation scale and driving strength. The intermediate half-step momentum is denoted by $\mathbf{p}_{t+\frac{\Delta t}{2}}$, and the gated leapfrog-style update is written as

$$\mathbf{p}_{t+\frac{\Delta t}{2}} = \mathbf{p}_t - \frac{g\,\Delta t}{2}\mathbf{F}(\mathbf{q}_t) \tag{14}$$

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$$\mathbf{q}_{t+\Delta t} = \mathbf{q}_t + \Delta t\, \mathbf{p}_{t+\frac{\Delta t}{2}} \tag{15}$$

$$\mathbf{p}_{t+\Delta t} = \mathbf{p}_{t+\frac{\Delta t}{2}} - \frac{g\,\Delta t}{2}\,\mathbf{F}\left(\mathbf{q}_{t+\Delta t}\right) \tag{16}$$

where g modulates the strength of the two momentum half-updates, while the position update retains the standard leapfrog form. T denotes the number of internal leapfrog-style evolution steps within each SHD block. In SSH-Net, T is shared across all SHD layers, while the projection parameters, learnable step size Δt, and gating coefficient g are separately instantiated and learned in each SHD layer.

To support the subsequent descriptor construction, the average of the two interaction responses is retained as the potential-field response tensor, denoted by $\overline{\mathbf{V}}_t$. This tensor summarizes the aggregated interaction-induced response within the current SHD layer and serves as the potential-field evidence used by the subsequent SSD.

Each SHD layer yields two outputs. The first is the evolution feature for subsequent discriminative representation learning. In particular, the updated position state is projected to the feature space and fused with the input feature through a residual connection followed by layer normalization, yielding

$$\mathbf{X}_{t+1} = \mathrm{LN}\left(\mathbf{q}_{t+\Delta t}\mathbf{W}_o + \mathbf{X}_t\right) \tag{17}$$

where $\mathbf{W}_o \in \mathbb{R}^{d\times d}$. The other is the phase-space state used for dynamical evidence readout

$$\mathcal{S}_t = \left(\mathbf{q}_{t+\Delta t}, \mathbf{p}_{t+\Delta t}, \overline{\mathbf{V}}_t\right). \tag{18}$$

Stacking L SHD layers yields an L-step discrete evolution process with layer index $t = 0, 1, \ldots, L-1$. The final layer, therefore, produces $\mathbf{X}_L$ and $\mathcal{S}_{L-1}$, where $\mathbf{X}_L$ is delivered to DEA as representation evidence and $\mathcal{S}_{L-1}$ is delivered to SSD as dynamical evidence.

2) Hamiltonian-Inspired Design Correspondence of SHD: The physical consistency of SHD is reflected in its structured evolution law. Following the Hamiltonian formulation in (3), SHD organizes spectral–spatial propagation with coupled latent position and latent momentum variables, so that feature updating follows an explicit evolution rule with a clear dynamic structure. In this way, the propagation process within each layer is grounded in a Hamiltonian-inspired dynamic template.

The potential-field response $\mathbf{V}(\mathbf{q})$ is also implemented as a tensorized interaction-induced response. It is generated by attention-style similarity and value aggregation, then aligned by a learnable linear mapping to act as the driving field in the update. Therefore, the difference from standard attention is not the absence of similarity weighting, but how the aggregated response is injected into a coupled position–momentum update instead of being used only as a one-step aggregated feature.

The update core of SHD follows the leapfrog-style transition in (14)–(16), where the driving response is evaluated before and after the position update. Together with the learnable parameters Δt and g, this organization provides a structured mechanism for modulating propagation scale and driving strength during feature evolution. Thus, SHD should be understood as a Hamiltonian-inspired, task-driven approximation of coupled feature propagation, without claiming strict physical symplectic conservation for the complete end-to-end classifier.

Algorithm 1 SHD Block

Input: token features X_t
Output: updated features X_{t+1} and state S_t

Linearly map X_t to position q_t, momentum p_t, key k, and value v
for $i = 1$ to T **do**
  $A \leftarrow \mathrm{softmax}\left(\frac{q_t k^\top}{\sqrt{d_k}}\right),\quad V^{(1)} \leftarrow Av$
  $p_{t+\frac{\Delta t}{2}} \leftarrow p_t - \frac{g\Delta t}{2}\mathrm{Linear}(V^{(1)})$
  $q_{t+\Delta t} \leftarrow q_t + \Delta t\, p_{t+\frac{\Delta t}{2}}$
  $A \leftarrow \mathrm{softmax}\left(\frac{q_{t+\Delta t} k^\top}{\sqrt{d_k}}\right),\quad V^{(2)} \leftarrow Av$
  $p_{t+\Delta t} \leftarrow p_{t+\frac{\Delta t}{2}} - \frac{g\Delta t}{2}\mathrm{Linear}(V^{(2)})$
  $V_t \leftarrow \frac{1}{2}(V^{(1)} + V^{(2)})$
end for
$X_{t+1} \leftarrow \mathrm{LN}(q_{t+\Delta t} W_o + X_t),\quad S_t \leftarrow (q_{t+\Delta t}, p_{t+\Delta t}, V_t)$

C. Symplectic State Descriptor

After L layers of SHD evolution, the final-layer state $\mathcal{S}_{L-1}$ is used as the input to SSD, where $\mathbf{q}_L, \mathbf{p}_L \in \mathbb{R}^{B\times N\times d}$, and $\overline{\mathbf{V}}_{L-1}$ denotes the final-layer potential-field response. These three components characterize the final representation distribution, propagation tendency, and interaction-induced driving, respectively. To convert such terminal dynamical evidence into a more compact and classification-oriented form, SSD condenses the final-layer state into a descriptor $\mathbf{d} \in \mathbb{R}^{B\times d_{\mathrm{desc}}}$ through statistical aggregation, lightweight projection, and joint fusion.

1) Position Descriptor: For the position state $\mathbf{q}_L \in \mathbb{R}^{B\times N\times d}$, first- and second-order statistics are extracted along the token dimension

$$\mu_q = \frac{1}{N}\sum_{i=1}^{N}\mathbf{q}_{L,i}, \qquad \sigma_q^2 = \mathrm{Var}\left(\mathbf{q}_L\right)\ \ (\text{over } i) \tag{19}$$

where $\mu_q \in \mathbb{R}^{B\times d}$ represents the channelwise mean of the final-layer token representations and reflects the global central state after evolution. $\sigma_q^2 \in \mathbb{R}^{B\times d}$ measures the channelwise dispersion of tokens, characterizing the compactness and heterogeneity of the final-state distribution. These two statistics provide a compact summary of the final-layer position distribution by capturing its central tendency and dispersion.

2) Momentum Descriptor: For the momentum state $\mathbf{p}_L \in \mathbb{R}^{B\times N\times d}$, the discriminative information is mainly reflected in the directional organization and retained strength of the final-layer propagation. Accordingly, to fully preserve the spatial heterogeneity in boundary regions and mixed pixels, four statistics are extracted to form a 4-D complementary momentum descriptor $\mathbf{p}_{desc} = [E_k, P_{mag}, P_{std}, P_{var}]$. First, the mean momentum vector is defined as

$$\bar{\mathbf{p}} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{p}_{L,i} \tag{20}$$

and its magnitude

$$P_{mag} = \|\bar{\mathbf{p}}\|_2 \tag{21}$$

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is used as a measure of the overall directional consistency of propagation. If the updating tendencies of different tokens are more aligned, P_{mag} becomes larger. If their updating directions cancel one another out, P_{mag} becomes smaller. Furthermore, the kinetic activity is defined as

$$E_k = \frac{1}{2N} \sum_{i=1}^{N} \left\| \mathbf{p}_{L,i} \right\|_2^2 \tag{22}$$

which characterizes the retained momentum strength in the final layer. Crucially, to prevent the smoothing of spatial variations, we explicitly extract the spatial standard deviation over the token dimension, denoted as

$$P_{std} = \mathrm{Std}_i \left(\left\| \mathbf{p}_{L,i} \right\|_2 \right). \tag{23}$$

This statistic acts as a dedicated sensor for spatial heterogeneity, yielding significantly higher values in boundaries due to divergent evolutionary tendencies among adjacent pixels. Finally, the channelwise variance $P_{var} = \mathrm{Var}_c(\bar{\mathbf{p}}_c)$ is computed. Together, these four statistics summarize both the macrolevel intensity and the fine-grained spatial dispersion of the terminal propagation.

3) Potential-Field Descriptor: The potential-field response $\overline{\mathbf{V}}_{L-1}$ originates from the aggregated interaction-driven term and reflects the integrated response of similarity-induced nonlocal interactions across head subspaces. Owing to its multihead structure, the head and channel dimensions are first merged, yielding the tensor $\widetilde{\mathbf{V}} \in \mathbb{R}^{B\times N\times (hd_v)}$. Statistics are then extracted along the token dimension

$$\mu_v = \frac{1}{N} \sum_{i=1}^{N} \widetilde{\mathbf{V}}_i, \qquad \sigma_v = \mathrm{Std}\left(\widetilde{\mathbf{V}}\right) \quad (\text{over } i). \tag{24}$$

Here, μ_v summarizes the average activation pattern of the multihead potential-field response over the entire patch and reflects the overall preference of nonlocal interaction. σ_v captures the fluctuation of driving strength across tokens and subspaces, describing the nonuniformity of interaction responses in boundary, heterogeneous, and category-mixed regions. Together, these statistics summarize the driving characteristics of the final interaction field.

4) Descriptor Fusion: After obtaining the above statistics, the three branch descriptors are formed and jointly fused as

$$\begin{aligned} d_q &= f_q\left(\left[\mu_q, \sigma_q^2\right]\right) \\ d_v &= f_v\left(\left[\mu_v, \sigma_v\right]\right) \\ d_p &= f_p\left(\left[E_k, P_{mag}, P_{std}, P_{var}\right]\right) \\ d &= f_{fus}\left(\left[d_q, d_p, d_v\right]\right). \end{aligned} \tag{25}$$

Here, $f_q(\cdot)$, $f_p(\cdot)$, $f_v(\cdot)$, and $f_{\mathrm{fus}}(\cdot)$ are lightweight MLPs with few parameters. The fusion stage jointly re-encodes the relations among position distribution, momentum tendency, and potential-field response in a low-dimensional space. The resulting $\mathbf{d}$ serves as token-length-invariant dynamical evidence that complements the backbone output. In this way, the descriptor branch condenses the analyzable phase-space structure of the final state into stable discriminative cues while retaining a direct connection to the classification objective. Since $\mathbf{d}$ participates in the classification loss during end-to-end training, it forms a constrained evidence channel integrated into the decision pathway.

D. DEA Branch

After L layers of SHD evolution, the final-layer feature sequence $\mathbf{X}_L \in \mathbb{R}^{B\times N\times d}$ encodes the representation obtained after structured dynamic propagation. Although this sequence already contains nonlocal interaction information, it does not explicitly summarize spatial statistics across different neighborhood ranges or compact patch-level semantic context. To complement the terminal dynamical evidence extracted by SSD, the DEA branch further organizes the final-layer representation through two components, namely multiscale spatial statistics and global context encoding.

The sequence feature $\mathbf{X}_L$ is first reshaped according to spatial topology into a feature map $\mathbf{M} \in \mathbb{R}^{B\times d\times P\times P}$, which restores the 2-D neighborhood structure of the patch and serves as the basis for subsequent spatial aggregation.

1) Multiscale Spatial Statistics: Within the DEA branch, four parallel pooling paths with kernel sizes 1, 3, 5, and 7 capture local-to-regional responses. Each path applies a convolution followed by global average pooling, producing vectors $\mathbf{u}_i$

$$\mathbf{u}_i = \mathrm{GAP}\left(\mathrm{Conv}_{k=i}\left(\mathbf{M}\right)\right), \qquad i \in \{1,3,5,7\}. \tag{26}$$

These vectors are concatenated to form the multiscale statistical feature $\mathbf{u} \in \mathbb{R}^{B\times d}$.

This component aggregates spatial statistics under different receptive fields with low parameter cost, enabling the final representation to preserve both fine local responses and broader regional summaries that are not explicitly encoded by sequence interaction alone.

2) Global Context Encoding: The second component summarizes the semantics of the entire patch by applying global average pooling to $\mathbf{M}$, flattening the pooled representation, and projecting it with a fully connected layer to obtain the contextual feature $\mathbf{g} \in \mathbb{R}^{B\times d/4}$. This branch provides a compact patch-level context that complements the multiscale spatial statistics with a more global semantic summary.

3) Evidence Aggregation and Classification: Finally, the multiscale statistical feature $\mathbf{u}$, the global contextual feature $\mathbf{g}$, and the SSD $\mathbf{d}$ are concatenated and fed to the classifier. Training uses the standard cross-entropy loss. In this aggregation, $\mathbf{u}$ provides spatial statistics across neighborhood ranges, $\mathbf{g}$ supplies compact global context, and $\mathbf{d}$ contributes terminal dynamical evidence from symplectic evolution. Their joint fusion forms the final discriminative representation used for prediction.

In terms of complexity, SHD similarity computation remains the dominant cost with order $\mathcal{O}(N^2)$, consistent with standard self-attention. The multiscale, global, and descriptor branches mainly involve pooling, statistical aggregation, and lightweight mappings whose cost grows approximately linearly with N, introducing limited additional overhead while improving the discriminative completeness of the final representation.

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TABLE I
CLASSIFICATION RESULTS ON THE INDIAN PINES DATASET. (1% TRAINING SAMPLES)

Class No.	CNN-Based		Transformer-Based			Mamba-Based		Attention-Augmented			Dynamics-Based	
	2D-CNN	HybridSN	SSFTT	MorphFormer	DSFormer	MambaHSI	MFormer	AMGCFN	MSNAT	CACFTNet	ODEnet	SSH-Net
1	1.86±1.74	56.36±11.54	31.82±31.53	58.18±18.33	62.79±20.96	88.37±13.64	86.22±14.43	82.79±15.61	22.79±5.58	44.19±22.79	63.11±17.30	72.27±24.20
2	76.16±11.07	56.89±9.44	79.20±14.88	78.73±8.66	78.58±6.29	83.08±3.10	76.28±8.89	82.07±4.92	77.46±6.48	79.97±6.16	82.15±3.73	82.50±6.71
3	55.18±9.22	65.31±3.48	69.90±12.02	67.12±14.89	77.16±9.42	72.41±16.48	78.66±6.08	90.10±4.19	77.34±5.68	66.61±11.83	77.37±1.65	81.75±7.58
4	8.94±7.54	27.64±8.29	70.39±23.59	75.50±18.77	76.64±6.93	74.34±19.80	73.16±8.30	64.42±23.25	82.04±6.00	71.77±14.96	83.68±6.24	87.71±3.06
5	67.82±3.06	42.45±5.63	67.70±11.09	74.93±4.39	76.07±15.56	87.39±9.45	84.39±3.47	75.16±8.02	69.33±8.00	78.06±22.32	68.33±14.14	78.90±6.32
6	94.40±2.52	90.17±1.74	91.99±6.31	93.53±3.12	95.77±3.15	97.74±2.54	96.45±3.06	94.37±2.79	92.46±1.79	94.26±3.72	91.50±7.00	94.82±2.41
7	13.08±19.40	66.15±12.02	86.92±18.76	86.15±27.69	95.38±5.65	100.00±0.00	100.00±0.00	100.00±0.00	82.31±22.51	85.38±13.16	99.26±1.48	99.23±1.54
8	91.00±7.34	99.06±1.88	98.12±3.75	98.46±1.57	96.03±5.27	78.82±15.88	97.17±3.39	99.87±0.26	98.51±1.05	94.89±4.48	99.70±0.25	96.84±3.19
9	2.22±4.44	86.67±10.30	35.56±35.92	45.56±31.70	78.89±8.89	71.11±36.41	93.68±10.21	95.56±8.89	40.00±20.61	94.44±6.80	92.63±9.18	94.44±8.61
10	46.35±4.15	66.68±7.53	69.14±13.19	71.01±8.81	72.62±6.69	79.96±4.98	78.81±4.49	81.46±6.73	80.75±3.17	77.73±4.41	81.27±5.44	81.83±3.95
11	80.75±10.42	87.98±2.82	86.25±5.44	88.48±3.90	85.73±2.10	88.61±3.71	86.70±1.41	92.44±4.05	87.67±2.91	87.95±2.79	92.25±4.18	92.77±2.63
12	38.77±19.57	45.99±9.78	42.62±17.18	53.32±11.77	47.28±6.31	68.26±14.21	59.18±10.65	46.01±13.65	40.77±7.61	54.17±17.54	66.03±13.30	63.99±10.31
13	95.69±4.63	65.07±4.54	80.50±18.87	94.97±7.89	94.46±5.56	90.87±6.10	90.00±8.34	98.15±1.64	97.54±2.74	91.49±6.87	94.26±4.11	94.37±5.20
14	99.01±1.11	89.99±3.03	92.56±5.80	92.88±1.80	97.27±1.29	96.85±1.73	98.42±0.81	94.30±4.07	85.81±2.73	98.39±0.63	98.75±1.15	98.27±1.10
15	35.57±10.43	39.42±1.33	61.59±16.17	74.71±11.95	63.46±18.20	78.27±13.71	68.06±14.61	88.92±6.00	77.03±5.38	57.84±11.94	72.30±8.56	78.57±12.53
16	5.17±7.64	21.10±9.84	72.97±35.01	88.57±10.96	95.96±3.38	73.03±32.14	82.61±19.72	82.02±12.71	96.40±5.57	99.10±1.46	89.57±4.49	91.21±5.65
OA (%)	71.03±1.91	71.64±1.19	78.80±3.39	81.52±2.31	81.86±2.18	84.87±1.92	83.84±0.89	86.20±1.70	81.15±1.51	82.26±2.02	86.00±1.96	**87.40±1.51**
AA (%)	59.59±6.73	62.93±2.10	71.08±7.82	77.63±4.83	80.88±2.88	83.07±3.20	84.36±1.21	85.48±2.65	75.51±2.85	79.76±2.27	84.51±2.95	**86.84±3.05**
κ (%)	66.36±2.12	67.31±1.62	75.75±3.99	78.83±2.73	79.19±2.53	82.75±2.17	81.53±1.01	84.30±1.91	78.50±1.75	79.67±2.38	84.01±2.25	**85.59±1.75**

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets

1) Indian Pines: Indian Pines is an airborne visible/infrared imaging spectrometer (AVIRIS) hyperspectral dataset over agricultural areas in Indiana, USA. The image size is 145×145 with 224 spectral bands (0.4–2.5 μm); after removing water-absorption bands, 200 bands are retained. It contains 16 crop-related classes with fragmented spatial distribution and frequent mixed pixels.

2) Houston2013: Houston2013 is an IEEE GRSS Data Fusion Contest dataset covering a complex urban scene. The image size is 349×1905 with 144 spectral bands (380–1050 nm) at 2.5-m spatial resolution. It includes 15 land-cover classes with diverse urban structures and strong interclass variability.

3) WHU-Hi-LongKou: WHU-Hi-LongKou, collected by Wuhan University, covers farmland scenes in Longkou, China. The image size is 550×400 with 270 spectral bands (0.4–1.0 μm), among which 224 bands are typically retained. The dataset contains nine land-cover classes and features clear parcel structures with complex boundary regions.

4) WHU-Hi-HanChuan: WHU-Hi-HanChuan is a UAV-borne hyperspectral dataset collected by Wuhan University over an agricultural scene in Hanchuan, Hubei, China. The image size is 1217×303 with 274 spectral bands covering approximately 400–1000 nm. It contains 16 crop and land-cover categories.

B. Experimental Settings

1) Implementation Details: All experiments are implemented in PyTorch on a workstation with an NVIDIA RTX 4090 GPU and an Intel Core i9 CPU. The patch size is set to 17×17 for WHU-Hi-HanChuan and to 13×13 for Indian Pines, Houston2013, and WHU-Hi-LongKou. For the SHD configuration, the number of stacked SHD layers is set to $L = 6$ for Indian Pines and $L = 3$ for Houston2013, WHU-Hi-LongKou, and WHU-Hi-HanChuan. The number of internal leapfrog-style evolution steps within each SHD block is shared across all SHD layers and set to $T = 1$. For optimization, all four datasets use AdamW with an initial learning rate of 1×10^{-3} together with weight decay and dropout regularization. All experiments are repeated five times, and the mean and standard deviation are reported.

2) Evaluation Metrics: The evaluation metrics include overall accuracy (OA), average accuracy (AA), and the kappa coefficient. OA measures the proportion of correctly classified test samples over all classes, AA computes the average of per-class accuracies, and kappa evaluates the agreement between predictions and ground truth beyond random chance.

3) Comparison Methods: We compare SSH-Net with eleven representative hyperspectral classification methods from four spectral–spatial modeling paradigms, including CNN-based models (2-D-CNN [44] and HybridSN [11]), Transformer-based models (SSFTT [13], MorphFormer [16], and DSFormer [17]), Mamba-based models (MambaHSI [23] and MFormer [22]), attention-augmented models (AMGCFN [36], MSNAT [37], and CACFTNet [38]), and dynamics-based model (ODEnet [43]). All comparison methods use publicly available implementations and recommended hyperparameters, and are trained under the same data splits, input settings, and training protocols for fair comparison.

C. Comparative Experimental Results and Analysis

Tables I–IV report the comparison results on the four datasets. Compared with CNN-, Transformer-, Mamba-, attention-augmented, and dynamics-based models, SSH-Net shows consistently competitive performance, and this result is related to how token interaction is organized through a Hamiltonian-inspired position–momentum update process. Most existing methods perform intralayer information mixing through one-shot similarity weighting, while cross-layer propagation mainly relies on stacking and residual connections. Dynamics-based models such as ODEnet reformulate feature propagation as continuous-depth evolution, but the interaction

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TABLE II

CLASSIFICATION RESULTS ON THE HOUSTON2013 DATASET. (1% TRAINING SAMPLES)

Class No.	CNN-Based		Transformer-Based			Mamba-Based		Attention-Augmented			Dynamics-Based	
	2D-CNN	HybridSN	SSFTT	MorphFormer	DSFormer	MambaHSI	MFormer	AMGCFN	MSNAT	CACFTNet	ODEnet	SSH-Net
1	91.98±4.74	85.63±8.04	87.42±5.85	90.30±4.97	92.65±4.93	85.77±3.30	89.21±5.04	89.81±4.61	91.33±4.44	92.62±5.98	90.06±6.33	94.33±5.48
2	91.57±4.70	91.87±7.40	95.54±4.41	91.48±4.61	93.75±6.32	93.95±4.31	92.51±4.93	87.12±5.89	93.27±5.15	93.84±6.95	94.00±5.89	92.36±4.55
3	99.91±0.18	91.33±4.46	98.75±0.86	96.60±0.78	94.93±1.07	98.21±2.34	97.25±1.48	98.72±1.18	98.83±1.16	99.30±0.68	98.43±1.19	97.91±1.85
4	95.26±3.00	89.08±11.37	96.83±2.01	93.68±2.43	98.08±2.63	95.88±1.82	92.10±2.76	88.27±10.43	93.40±3.36	97.03±3.55	92.85±3.00	94.65±1.71
5	99.90±0.20	99.36±0.70	99.95±0.07	99.51±0.47	98.67±1.37	99.46±0.73	99.90±0.12	98.45±1.97	99.36±0.64	97.75±4.55	96.91±5.82	99.24±1.45
6	92.86±5.57	65.17±24.86	81.68±7.46	75.65±11.33	89.10±4.23	84.05±6.69	82.37±6.80	85.23±7.28	86.75±3.85	84.16±8.54	75.45±4.83	81.62±6.06
7	75.64±5.08	79.44±4.75	87.03±4.43	88.57±3.15	93.29±2.82	88.63±4.63	84.69±4.39	90.01±1.46	84.90±4.83	79.73±7.40	85.16±6.17	94.47±3.81
8	65.63±5.14	63.15±4.64	77.58±5.61	62.50±5.00	71.97±9.57	63.76±3.91	63.36±7.31	65.25±4.61	71.03±4.73	68.75±7.80	62.01±6.84	70.87±7.37
9	53.62±19.03	69.07±7.09	77.87±2.99	81.47±5.37	84.50±1.58	85.35±2.49	77.37±4.51	75.80±7.26	82.19±6.10	85.37±3.91	84.99±6.70	87.51±4.46
10	68.12±16.82	81.58±4.32	77.73±12.49	92.27±5.63	83.64±6.99	67.49±7.08	90.20±9.90	94.71±7.04	82.58±7.94	82.95±10.07	76.82±20.21	95.77±2.91
11	62.75±14.33	76.39±13.71	81.57±6.80	81.21±9.48	83.08±10.78	82.69±9.39	89.79±10.00	92.70±7.66	85.56±6.54	82.70±8.64	80.56±11.57	83.68±9.76
12	62.54±18.12	87.41±4.73	84.39±6.88	89.98±5.94	72.90±14.99	71.48±6.39	83.61±10.11	79.72±5.49	86.08±6.33	72.99±16.02	73.64±14.25	87.03±6.73
13	32.07±24.97	38.82±16.01	89.61±3.37	87.32±4.74	94.44±1.01	78.49±7.24	87.28±2.72	86.77±2.26	81.61±5.33	83.22±8.49	89.01±8.22	92.03±2.22
14	99.02±0.93	97.94±2.56	100.00±0.00	98.61±1.45	99.01±0.72	99.37±0.87	99.57±0.46	99.86±0.19	96.79±2.33	99.43±1.15	97.97±1.71	99.57±0.53
15	100.00±0.00	99.69±0.39	98.32±3.22	94.95±5.83	100.00±0.00	99.81±0.18	97.70±4.59	97.49±4.95	96.32±3.89	99.81±0.25	99.91±0.12	99.60±0.80
OA (%)	78.42±0.56	82.20±2.52	88.05±1.11	87.97±0.69	88.80±1.63	85.20±2.21	87.60±1.30	87.64±0.93	88.04±1.31	86.97±1.46	85.52±2.60	**90.94±1.44**
AA (%)	79.39±1.28	81.06±3.05	88.95±1.05	88.27±1.19	90.00±1.14	86.29±2.07	88.46±0.92	88.66±0.74	88.67±1.36	87.98±1.62	86.52±2.02	**91.38±1.19**
κ (%)	76.66±0.61	80.73±2.73	87.08±1.20	86.99±0.75	87.89±1.76	84.00±2.39	86.60±1.40	86.65±1.00	87.07±1.42	85.91±1.58	84.34±2.81	**90.21±1.56**

TABLE III

CLASSIFICATION RESULTS ON THE WHU-HI-LONGKOU DATASET. (0.1% TRAINING SAMPLES)

Class No.	CNN-Based		Transformer-Based			Mamba-Based		Attention-Augmented			Dynamics-Based	
	2D-CNN	HybridSN	SSFTT	MorphFormer	DSFormer	MambaHSI	MFormer	AMGCFN	MSNAT	CACFTNet	ODEnet	SSH-Net
1	99.51±0.38	99.75±0.27	99.89±0.09	99.43±0.37	99.87±0.13	99.57±0.39	99.83±0.15	99.39±0.32	99.32±0.82	98.83±0.98	99.73±0.18	99.85±0.05
2	74.46±5.55	83.57±9.69	93.10±2.99	97.19±0.89	89.40±7.17	99.20±0.50	88.99±7.82	95.97±2.93	87.28±10.11	90.18±8.16	83.84±9.87	98.89±0.73
3	47.17±33.53	86.39±5.54	80.54±12.49	88.95±5.16	94.85±2.62	97.04±1.82	84.59±3.28	96.33±2.50	84.78±9.31	85.26±4.44	90.47±11.38	92.53±6.56
4	97.85±0.71	97.31±1.02	98.74±0.28	99.28±0.39	97.56±0.41	98.27±1.24	98.78±0.34	98.66±0.24	98.41±0.63	98.55±0.83	98.56±0.49	99.43±0.07
5	64.92±12.21	66.34±24.11	44.86±16.74	88.79±1.72	84.78±13.01	82.46±11.76	85.57±12.02	96.25±2.03	85.68±5.99	70.90±15.12	65.70±14.41	91.20±6.11
6	93.43±4.71	97.29±2.35	98.71±1.52	92.61±1.68	97.31±2.72	90.89±3.44	96.51±4.05	94.20±3.54	95.35±2.74	96.22±3.83	93.73±8.00	98.82±0.42
7	99.97±0.02	99.95±0.07	99.94±0.07	99.88±0.11	99.98±0.02	99.93±0.07	98.19±1.10	99.71±0.20	99.75±0.36	99.81±0.21	99.73±0.16	99.26±0.60
8	82.21±6.25	91.70±2.38	90.00±4.11	90.06±1.44	92.92±1.68	81.30±5.57	82.98±3.72	77.85±13.67	85.73±6.18	93.44±3.52	87.67±8.10	92.81±4.34
9	56.72±26.92	78.43±4.39	72.60±8.15	72.67±10.37	85.96±3.54	72.55±9.81	76.71±9.67	82.88±5.67	80.99±9.45	67.67±6.71	74.43±9.52	84.53±7.52
OA (%)	94.60±0.59	96.55±0.92	96.76±0.33	97.66±0.19	97.64±0.48	97.06±0.47	96.64±0.87	97.55±0.37	97.02±1.03	96.81±0.61	96.48±0.90	**98.51±0.65**
AA (%)	79.58±3.91	88.97±4.24	86.49±2.09	92.10±1.30	93.63±1.57	91.24±1.30	90.24±2.05	93.47±1.47	90.81±2.60	88.98±2.56	88.21±3.01	**95.26±2.28**
κ (%)	92.83±0.79	95.46±1.22	95.73±0.43	96.92±0.25	96.89±0.63	96.12±0.62	95.58±1.15	96.77±0.48	96.08±1.35	95.79±0.81	95.36±1.19	**98.04±0.85**

TABLE IV

CLASSIFICATION RESULTS ON THE WHU-HI-HANCHUAN DATASET. (1% TRAINING SAMPLES)

Class No.	CNN-Based		Transformer-Based			Mamba-Based		Attention-Augmented			Dynamics-Based	
	2D-CNN	HybridSN	SSFTT	MorphFormer	DSFormer	MambaHSI	MFormer	AMGCFN	MSNAT	CACFTNet	ODEnet	SSH-Net
1	94.43±0.64	98.22±0.58	97.65±0.77	98.44±0.69	98.41±0.57	99.63±0.11	98.98±0.38	99.38±0.08	97.85±0.57	97.87±0.24	97.95±2.47	99.60±0.14
2	89.57±0.51	94.52±1.56	95.76±2.48	96.98±1.75	96.19±1.04	98.30±0.29	98.00±0.36	97.59±0.57	95.97±1.46	81.34±9.33	97.86±0.83	99.46±0.16
3	70.53±5.59	94.99±2.63	98.59±1.14	96.76±0.95	94.51±4.59	99.41±0.22	96.58±1.51	99.19±0.71	93.05±4.93	83.71±11.10	97.50±2.17	99.60±0.25
4	80.73±8.18	96.82±1.27	97.07±1.39	97.77±0.38	98.03±0.85	99.12±0.27	99.15±0.28	97.45±1.38	95.51±2.95	52.68±44.92	98.76±1.35	99.63±0.25
5	10.64±6.22	65.02±12.95	67.96±21.49	88.21±5.85	79.98±6.05	97.64±1.51	88.86±4.32	86.35±7.81	59.90±32.95	69.34±10.03	97.24±3.21	97.58±2.43
6	15.61±6.12	57.10±18.51	75.49±8.25	79.70±3.04	68.42±15.71	89.04±3.49	84.93±5.46	90.15±3.92	63.52±16.19	53.38±7.98	77.79±5.62	89.54±3.84
7	65.07±4.24	88.30±5.00	90.89±3.96	93.82±2.61	91.01±3.05	95.53±2.78	94.63±0.94	97.51±0.29	88.12±3.86	91.87±1.35	94.92±3.79	97.59±1.33
8	73.02±2.25	86.90±5.03	89.22±4.68	94.09±1.92	89.24±11.11	97.57±0.77	95.46±1.89	96.27±1.05	86.95±5.70	76.82±11.11	86.85±4.17	97.98±1.44
9	47.76±5.51	87.90±2.83	92.36±5.66	93.61±3.70	92.20±2.08	93.55±2.29	95.46±1.08	93.23±2.98	84.54±10.44	82.28±4.02	93.40±5.11	97.65±0.51
10	76.07±5.40	98.68±0.89	96.77±1.75	98.07±0.50	98.63±0.84	98.08±1.08	98.82±0.16	95.77±1.63	97.34±1.22	95.16±1.69	98.17±1.44	98.91±0.51
11	91.44±3.82	95.58±2.53	96.62±1.89	96.69±0.90	96.35±1.35	99.27±0.17	97.82±0.79	98.62±0.48	94.83±3.02	97.38±0.56	95.39±3.33	99.17±0.13
12	1.64±0.70	72.33±8.35	74.40±12.08	90.79±3.28	84.47±3.05	96.62±2.99	90.81±1.66	99.27±1.46	59.20±30.11	81.07±4.94	91.54±5.31	99.01±0.77
13	46.34±5.99	74.57±3.39	82.47±5.78	82.51±3.03	80.42±3.78	92.49±1.76	87.22±1.47	86.04±2.07	74.75±12.50	78.83±2.87	88.95±2.49	93.64±0.31
14	82.72±1.94	93.66±1.23	94.02±3.32	94.61±0.57	93.74±3.02	96.43±0.59	96.11±1.27	95.76±0.50	93.65±3.85	62.80±11.39	90.91±6.06	98.49±0.70
15	76.41±7.27	66.10±8.72	74.15±7.62	84.42±7.67	76.07±9.99	73.16±12.52	75.61±8.04	67.10±2.37	67.84±12.75	7.89±6.87	81.99±8.91	82.54±5.63
16	98.30±0.31	99.81±0.15	99.79±0.06	99.71±0.09	99.25±0.76	99.85±0.11	99.80±0.14	99.69±0.18	99.58±0.31	99.62±0.19	96.42±6.48	99.88±0.05
OA (%)	83.34±0.28	94.09±0.60	95.33±0.85	96.50±0.26	95.25±1.13	98.17±0.11	97.35±0.18	97.48±0.19	93.67±2.67	88.34±2.06	94.98±3.30	**98.82±0.08**
AA (%)	63.77±0.64	85.65±2.27	88.95±3.09	92.89±0.39	89.80±1.96	95.36±0.71	93.64±0.24	93.71±0.50	84.54±7.44	75.75±4.40	92.85±2.16	**96.89±0.48**
κ (%)	80.47±0.35	93.07±0.70	94.53±1.00	95.90±0.30	94.44±1.32	97.86±0.13	96.89±0.21	97.05±0.22	92.58±3.14	86.31±2.42	94.16±3.79	**98.62±0.10**

is still governed by unconstrained vector-field updates. In SSH-Net, the similarity-induced interaction response continuously participates in state updating, and cross-layer propagation is guided by a structured leapfrog-style update. As a result, feature mixing is accumulated progressively across layers rather than produced as a static output within a single layer. The comparative results across the four datasets are aligned with this more coherently organized updating mechanism.

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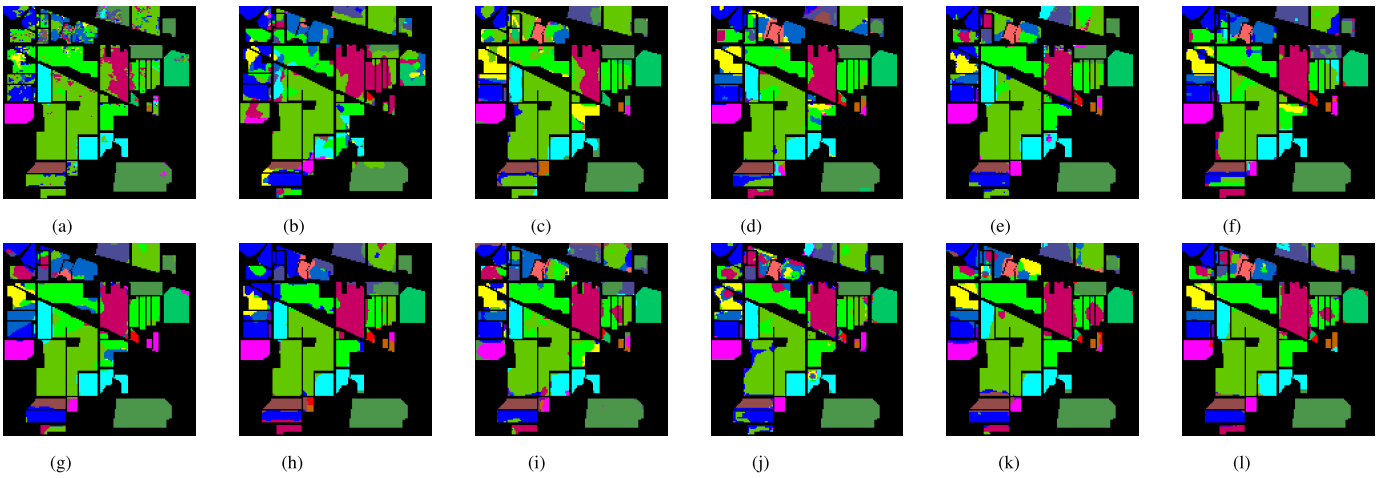


Fig. 3. Classification maps of different models on the Indian Pines dataset with 1% samples per class as the training set. (a) 2D-CNN. (b) HybridSN. (c) SSFTT. (d) MorphFormer. (e) DSFormer. (f) MambaHSI. (g) MFormer. (h) AMGCFN. (i) MSNAT. (j) CACFTNet. (k) ODEnet. (l) SSH-Net.

In Table I, SSH-Net achieves the highest OA, AA, and κ on Indian Pines, reaching 87.40%, 86.84%, and 85.59%, respectively. This gain is closely related to the SHD updating scheme. Indian Pines contains many mixed pixels and highly similar classes. On Indian Pines, many confusing categories exhibit high spectral similarity and strong interclass ambiguity, making fine-grained discrimination particularly challenging. Compared with one-step interaction models, SSH-Net introduces a structured multistep update mechanism in which the learned interaction response is incorporated through two consecutive leapfrog-style evaluations within each layer. Empirically, this design is associated with better discrimination among several confusing classes, suggesting that organized multistep propagation may help reduce the accumulation of local spectral perturbations across layers.

In Table II, SSH-Net attains 90.94% OA, 91.38% AA, and 90.21% κ on Houston2013. This dataset involves large-scale variation, linear structures, boundary regions, and sparse targets, which place higher demands on both local detail modeling and broader spatial context. In SSH-Net, the final-layer features are rearranged into a spatial feature map, and statistics from multiple receptive fields together with global contextual encoding are incorporated through DEA. Empirically, this state-aware aggregation strategy appears beneficial for structurally complex scenes, where relying only on token-sequence interaction outputs may be insufficient to capture both local and regional discriminative cues.

In Table IV, SSH-Net achieves the best results on WHU-Hi-HanChuan in terms of OA, AA, and κ. This dataset contains fragmented field boundaries and heterogeneous agricultural parcels, where local spectral variations can easily lead to scattered predictions. The advantage of SSH-Net may benefit from the terminal-state evidence aggregation in DEA. By integrating terminal-state statistics with multiscale spatial and global contextual evidence, DEA provides a more stable decision basis for heterogeneous parcels and boundary regions.

Figs. 3–6 present the classification maps. Compared with the competing methods, SSH-Net tends to produce fewer scattered errors within homogeneous regions and better continuity near narrow strips, intersections, and complex boundaries. These visual results are qualitatively consistent with the quantitative improvements reported in OA, AA, and κ. On WHU-Hi-HanChuan, the classification maps further show that SSH-Net preserves large agricultural parcels while reducing fragmented false predictions in heterogeneous areas.

Fig. 7 provides consistent evidence from the embedding space. Compared with HybridSN, DSFormer, MambaHSI, AMGFCN, and ODEnet, SSH-Net exhibits overall clearer classwise organization and relatively more compact intraclass distributions. The difference between SSH-Net and ODEnet is not large at the global level, since both methods already produce fairly separable class clusters. However, SSH-Net still shows slightly better compactness and cluster coherence in several categories, indicating a modest advantage in preserving discriminative structure during feature propagation. This difference is related to how the final-state evidence is organized. SSH-Net does not rely only on the final-layer backbone features. It further introduces three readout branches, namely multiscale spatial statistics, global context, and final-state dynamical descriptors, so that classification uses local structure, global semantics, and evolved state information jointly. Meanwhile, SHD organizes interaction through a structured coupled update, while SSD and DEA improve the discriminative readability of the final-state information, leading to clearer class distributions in the embedding space.

D. Ablation Studies and Analysis

1) Ablation Study of Model Components: As shown in Table V, we perform component-wise ablation on Indian Pines. Removing the symplectic dynamics causes clear drops of 3.39, 5.50, and 3.91 percentage points in OA, AA, and κ, respectively, supporting that the coupled dynamic update contributes beyond deeper static interaction stacking. The descriptor branch also plays a substantial role. Removing SSD reduces OA, AA, and κ by 1.76, 3.75, and 2.02 percentage points, showing that backbone propagation alone cannot fully expose the discriminative information accumulated in the evolved terminal state. Replacing SSD with a simple

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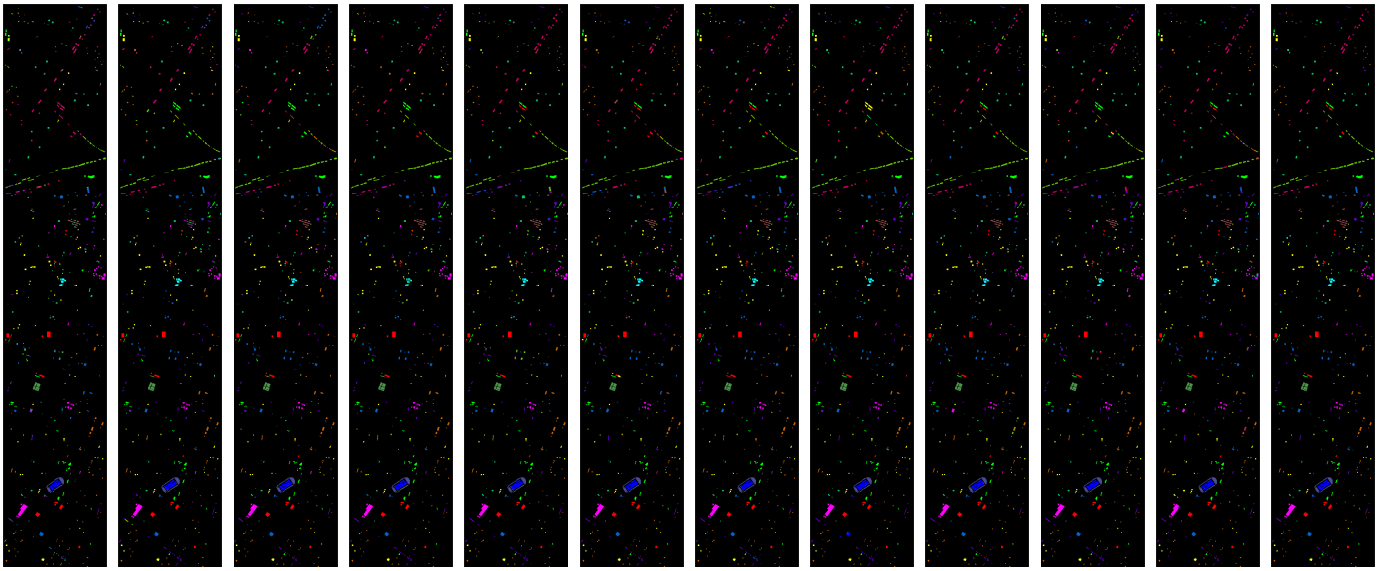


Fig. 4. Classification maps of different models on the Houston2013 dataset with 1% samples per class as the training set. (a) 2D-CNN. (b) HybridSN. (c) SSFTT. (d) MorphFormer. (e) DSFormer. (f) MambaHSI. (g) MFormer. (h) AMGCFN. (i) MSNAT. (j) CACFTNet. (k) ODEnet. (l) SSH-Net.

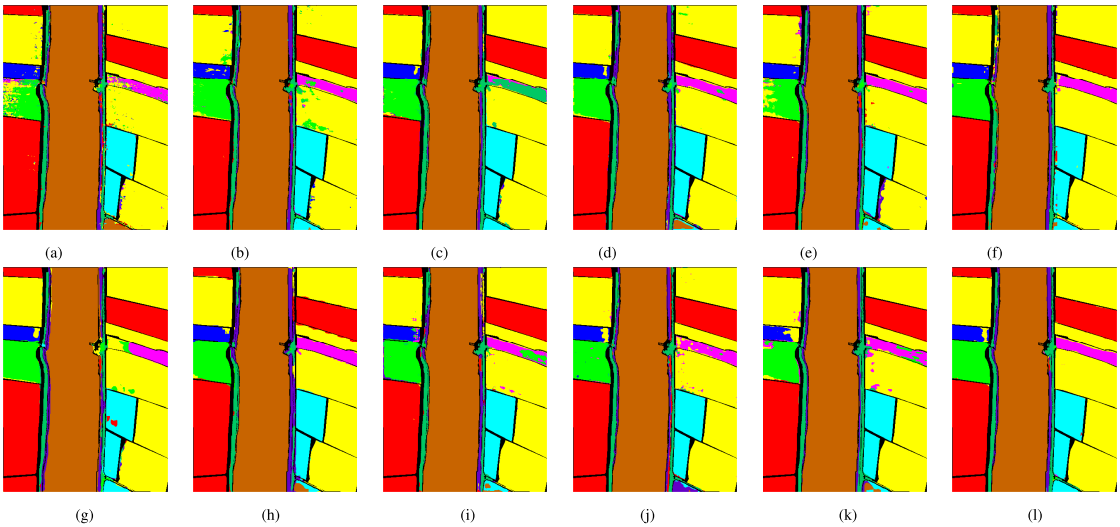


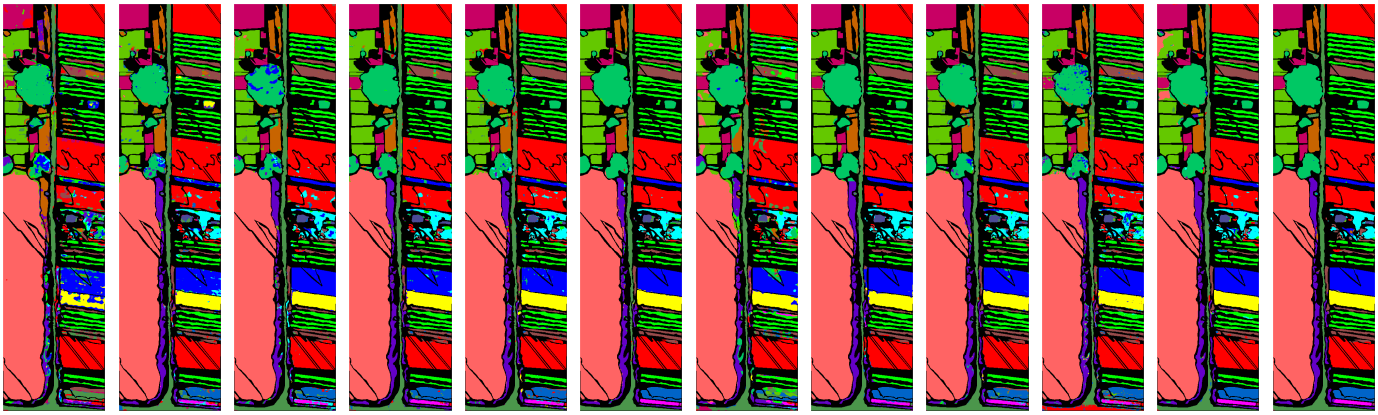
Fig. 5. Classification maps of different models on the WHU-Hi-LongKou dataset with 0.1% samples per class as the training set. (a) 2D-CNN. (b) HybridSN. (c) SSFTT. (d) MorphFormer. (e) DSFormer. (f) MambaHSI. (g) MFormer. (h) AMGCFN. (i) MSNAT. (j) CACFTNet. (k) ODEnet. (l) SSH-Net.

global average pooling readout still leads to declines of 1.39, 2.48, and 1.59 percentage points, respectively. This comparison indicates that the gain is not merely due to introducing an additional terminal-state branch, but to the more effective organization and readout of dynamical evidence by SSD.

The remaining two components mainly provide auxiliary support for the final decision. Removing global context encoding causes relatively mild but consistent decreases of 0.71, 1.41, and 0.83 percentage points, suggesting that scene-level context mainly acts as a stabilizing complement. Removing multiscale fusion leads to larger drops of 2.73, 4.33, and 3.13 percentage points, indicating that scale-aligned spatial aggregation remains important when terminal dynamical evidence is converted into class predictions. Overall, the ablation results show that SSH-Net benefits primarily from the coupling of structured symplectic evolution and explicit terminal-state readout, while global context and multiscale fusion further improve the completeness of the final evidence.

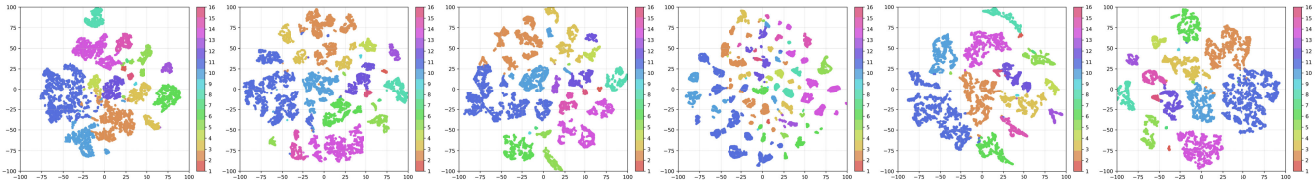
2) Grad-CAM Visualization Under Component Ablation: Fig. 8 presents scene-level Grad-CAM results under shared global normalization. The full SSH-Net produces the clearest and most spatially coherent response, with activation distributed more continuously over the target structures. Removing the symplectic dynamics weakens this response and makes it more diffuse, indicating that structured dynamical interaction is more effective than static attention aggregation for maintaining consistent feature propagation. Without SSD, the activation becomes more scattered and extends into surrounding irrelevant regions, suggesting that this branch

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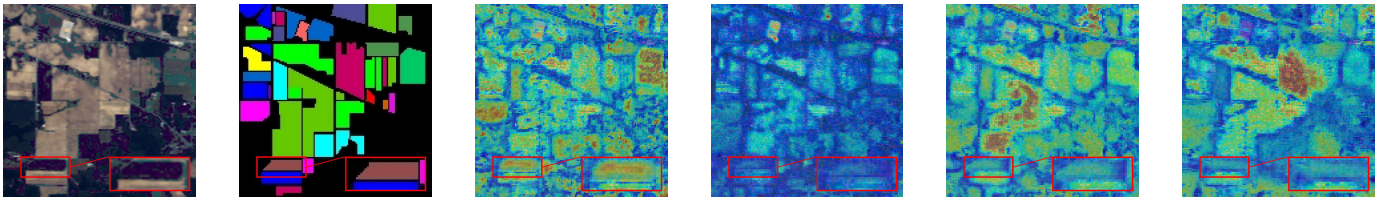
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 6. Classification maps of different models on the WHU-Hi-HanChuan dataset with 1% samples per class as the training set. (a) 2D-CNN. (b) HybridSN. (c) SSFTT. (d) MorphFormer. (e) DSFormer. (f) MambaHSI. (g) MFormer. (h) AMGCFN. (i) MSNAT. (j) CACFT-Net. (k) ODEnet. (l) SSH-Net.



(a) (b) (c) (d) (e) (f)

Fig. 7. t-SNE visualization of the feature embeddings on the Indian Pines dataset with 10% training samples per class. (a) HybridSN. (b) DSFormer. (c) MambaHSI. (d) AMGCFN. (e) ODEnet. (f) SSH-Net.



(a) (b) (c) (d) (e) (f)

Fig. 8. Scene-level Grad-CAM visualization under component ablation. All Grad-CAM results are generated under shared global normalization. (a) RGB image. (b) Ground truth. (c) SSH-Net. (d) w/o symplectic dynamics. (e) w/o SSD. (f) w/o DEA.

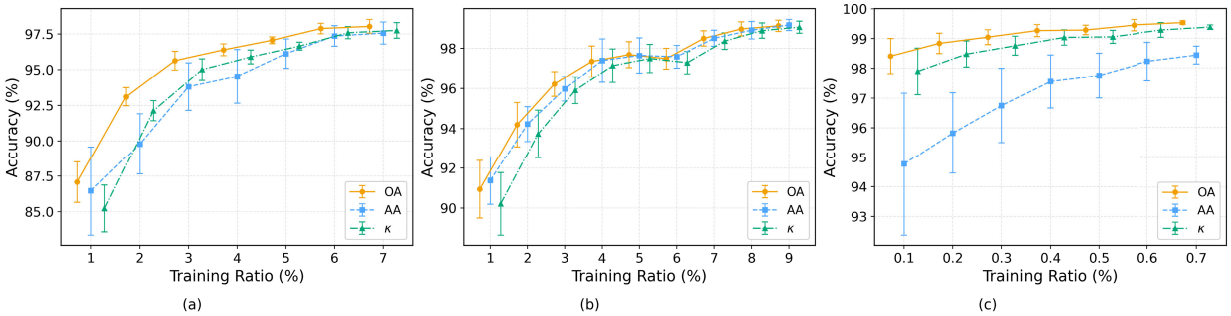


Fig. 9. Performance trends under different training ratios on (a) Indian Pines, (b) Houston2013, and (c) WHU-Hi-LongKou.

improves the readability of discriminative evidence. Without DEA, the response is biased toward a few local areas and no longer supports complete structure-level localization. These visual patterns are in line with the quantitative ablation results and further show that SSH-Net yields more coherent and spatially consistent evidence for classification.

3) Ablation Study of Symplectic-Inspired Dynamic Update and Evolution Depth: To evaluate the role of SHD at the interaction-paradigm level, we conduct an ablation study on the symplectic-inspired dynamic update mechanism. SHD uses attention-style similarity to construct an interaction response, then injects this response into the position–momentum

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TABLE V

COMPONENT ABLATION RESULTS ON INDIAN PINES. "–" DENOTES REMOVING THE CORRESPONDING COMPONENT

Component	OA (%)	AA (%)	κ (%)
Full model	**87.44±1.32**	**86.37±2.28**	**85.67±1.50**
– Symplectic dynamics	84.05±1.48	80.87±2.85	81.76±1.70
– Descriptor branch	85.68±1.47	82.62±2.40	83.65±1.68
SSD replaced by GAP	86.05±2.01	83.89±2.58	84.08±2.29
– Global context encoding	86.73±0.89	84.96±2.09	84.84±1.01
– Multi-scale fusion	84.71±0.46	82.04±1.32	82.54±0.56

TABLE VI

STUDY ON SYMPLECTIC-INSPIRED DYNAMIC UPDATE VERSUS STATIC ATTENTION ON INDIAN PINES WITH L STACKED INTERACTION LAYERS

L	Dynamic (Evolution)			Static (Attention Stacking)		
	OA	AA	κ	OA	AA	κ
1	87.24±1.25	86.19±1.69	85.45±1.43	84.57±0.93	82.50±1.43	82.36±1.07
2	86.23±1.92	85.88±2.62	84.28±2.19	84.22±1.95	80.12±4.44	81.94±2.23
3	86.54±2.06	85.79±2.76	84.65±2.33	84.29±1.52	82.67±3.23	82.03±1.78
4	86.44±2.38	86.00±2.63	84.53±2.70	83.07±2.06	82.38±2.66	80.64±2.39
5	86.41±0.82	85.21±1.68	84.50±0.92	83.54±1.33	83.55±1.96	81.24±1.47
6	87.26±1.28	86.39±2.32	85.47±1.46	84.05±1.48	80.87±2.85	81.76±1.70
7	86.96±0.80	86.51±0.53	85.12±0.92	84.26±1.63	80.86±2.10	82.02±1.88
8	86.57±1.41	85.39±1.39	84.70±1.59	84.31±1.63	82.12±1.91	82.06±1.87
9	86.22±1.50	85.29±2.56	84.29±1.73	83.38±1.45	80.21±3.67	81.02±1.65
10	86.62±0.84	85.91±0.93	84.75±0.96	84.04±1.70	82.51±2.16	81.79±1.95

coupled update through a learned driving field. Accordingly, we construct a depth- and training-matched static baseline using standard self-attention with one-step aggregation. Since only the interaction mechanism is changed, the performance gap mainly reflects the effect of the update form.

As shown in Table VI, the dynamic update model consistently outperforms the static baseline over depths from $L = 1$ to 10 and is less sensitive to depth variation. On average, dynamic interaction improves OA, AA, and κ by 2.68%, 4.08%, and 3.09%, respectively. This difference suggests more stable cross-layer propagation in hyperspectral classification, where noise, mixed pixels, and boundary uncertainty make static attention more prone to oversmoothing and routing bias. By comparison, SHD uses similarity-induced interaction as the driving signal, organizes updates through latent position and momentum states, and employs a leapfrog-style update to make propagation more controllable and consistent.

The best performance appears at moderate depth rather than increasing monotonically, which is consistent with discrete-time evolution: too few steps are insufficient for nonlocal propagation, whereas too many yield diminishing returns and may introduce slight oscillation. This indicates that the advantage of SHD lies in stable information diffusion and structural alignment within a proper number of steps. Overall, the gains are associated with latent phase-space decomposition, coupled state updating, and explicit regulation of propagation intensity.

4) Ablation Study of Evolution Steps: To evaluate the proposed symplectic-inspired dynamic update within each SHD block, we vary only the shared number of evolution steps T while keeping all other settings unchanged. As shown in Table VII, the model remains stable for $T = 1 \sim 3$,

TABLE VII

EFFECT OF SHARED EVOLUTION STEPS T ON OA/AA/κ

T	Indian Pines (%)			Houston2013 (%)			WHU-Hi-LongKou (%)		
	OA	AA	κ	OA	AA	κ	OA	AA	κ
1	**87.40±1.51**	**86.84±3.05**	**85.59±1.75**	**90.94±1.44**	91.38±1.10	**90.21±1.56**	98.51±0.65	**95.26±2.28**	98.04±0.85
2	87.06±1.53	85.82±2.80	85.24±1.75	90.67±1.81	**91.43±1.44**	89.91±1.96	98.52±0.45	94.65±2.19	98.05±0.60
3	87.35±1.25	85.11±2.43	85.57±1.42	90.69±1.27	91.41±1.02	89.94±1.37	**98.67±0.40**	95.21±1.41	**98.25±0.52**
4	86.68±1.32	84.69±2.44	84.82±1.50	90.75±1.66	91.35±1.39	90.00±1.80	98.61±0.58	94.83±2.66	98.17±0.76

TABLE VIII

EFFECT OF PATCH SIZE ON OA/AA/κ

Patch	Indian Pines (%)			Houston2013 (%)			WHU-Hi-LongKou (%)		
	OA	AA	κ	OA	AA	κ	OA	AA	κ
5	74.34±2.63	68.65±2.56	70.59±3.04	86.94±1.14	88.40±0.93	85.88±1.23	96.78±0.47	88.94±1.74	95.76±0.62
7	80.15±2.45	77.52±3.12	77.24±2.85	88.58±1.20	89.94±0.72	87.65±1.29	97.89±0.53	93.54±2.12	97.23±0.70
9	83.88±3.14	81.45±4.66	81.53±3.59	88.48±1.80	89.64±1.39	87.55±1.95	98.09±0.57	93.80±2.89	97.49±0.75
11	86.50±1.35	84.67±2.02	84.57±1.58	90.89±1.90	**91.61±1.54**	90.16±2.05	98.30±0.47	94.51±2.33	97.76±0.63
13	**87.40±1.51**	**86.84±3.05**	**85.59±1.75**	**90.94±1.44**	91.38±1.19	**90.21±1.56**	**98.51±0.65**	**95.26±2.28**	**98.04±0.85**
15	86.61±1.67	85.52±1.96	84.72±1.89	89.53±1.37	90.04±1.06	88.68±1.48	98.40±0.36	94.46±2.01	97.89±0.48
17	86.66±1.25	85.16±2.52	84.79±1.38	88.75±1.29	89.24±1.00	87.83±1.39	98.25±0.39	94.18±1.60	97.69±0.51

suggesting that a few structured propagation steps are sufficient to complete the main spectral–spatial interaction and state reorganization for classification. This is consistent with the design goal of achieving efficient information propagation through regulated latent phase-space evolution rather than unstructured long-chain iteration.

When T increases to 4, the performance declines, especially in AA, indicating that excessive evolution tends to smooth cross-class responses and weaken the separability of boundary and minority classes. Since interaction weights are re-estimated during state updating, a larger T corresponds to a longer propagation chain and stronger information mixing, which also increases cross-class coupling. Considering both performance and cost, a smaller number of evolution steps better preserves propagation consistency while avoiding degradation from excessive propagation.

5) Ablation Study of Training Ratio: Fig. 9 shows that performance improves steadily with the training ratio, especially in the lower training-ratio regime, indicating that SHD mainly benefits from its explicitly organized interaction and update mechanism rather than relying only on higher training ratios. By organizing propagation through a regulated position–momentum update, SHD helps the model form effective discriminative structures at different training ratios. As the training ratio increases further, the gains gradually saturate, with later improvements mainly reflected in boundary refinement and bias reduction. Overall, the proposed method maintains stable performance trends across different training ratios.

6) Ablation Study of Patch Size: Table VIII shows that patch size affects performance by setting the initial phase-space interaction range of SHD. Small windows provide limited spatial context, making stable region-level evolution difficult. Performance is best at 11–13, where cross-pixel interaction, state propagation, and terminal-state readout are better coordinated. Larger windows introduce more heterogeneous regions, increase state mixing, and weaken terminal discriminative evidence, causing saturation or decline. Overall, patch size determines the phase-space modeling boundary and the quality of structured propagation in SHD.

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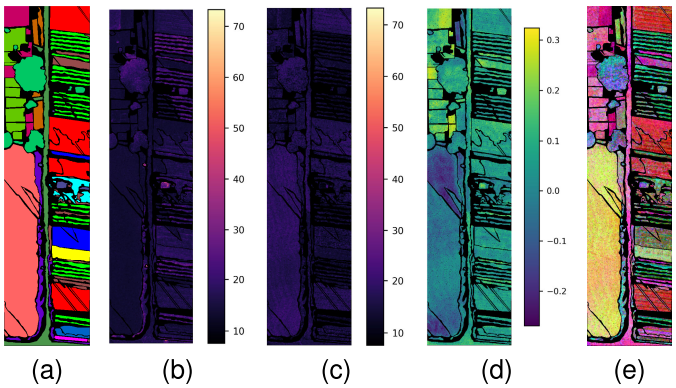


Fig. 10. Visualization of the learned terminal phase-space states on the WHU-Hi-HanChuan dataset. (a) Ground truth map, (b) kinetic energy E_k at Layer 1, (c) kinetic energy E_k at the last layer, showing attenuated momentum responses near heterogeneous transitions, (d) potential-field response V at the last layer, showing a spatially nonuniform interaction-response pattern, and (e) terminal position state q visualized by 3-D PCA pseudocolor, showing class-consistent spatial organization and visible boundary structures.

7) Visualization of Learned Phase-Space States: To qualitatively examine the learned latent phase-space behavior, we visualize the position state q, the momentum-related kinetic energy E_k, and the potential-field response V on WHU-Hi-HanChuan, whose fragmented boundaries and heterogeneous land covers provide a challenging scene-level case.

As shown in Fig. 10, the first-layer kinetic energy is relatively scattered over textured regions, whereas the last-layer map exhibits more organized responses and lower values along many heterogeneous transitions. This attenuation near boundaries suggests that momentum responses tend to be weakened where neighboring tokens receive inconsistent interaction cues, which is consistent with reduced cross-class mixing during propagation. The terminal potential-field response also shows a spatially nonuniform pattern, with higher activations around local structures and transition regions, suggesting its role as an interaction-driven guide for position–momentum updating.

The terminal position state q, visualized by 3-D principal component analysis (PCA) pseudocolor, presents class-consistent spatial organization broadly aligned with the ground truth. Large homogeneous regions remain continuous, while many fragmented boundaries are still visible, providing qualitative support that the proposed update helps preserve spatial heterogeneity in the terminal feature state.

VI. Conclusion

In this article, we proposed SSH-Net, a symplectic-inspired dynamic interaction network for HSIC. The proposed framework reformulates spectral–spatial token interaction from one-step aggregation into a structured multistep propagation process by modeling token representations as latent position and momentum states and updating them with gated leapfrog-style transitions. Based on this dynamic backbone, SSH-Net further incorporates an SSD branch to summarize terminal latent-state statistics as compact evidence, and a DEA module to combine such evidence with multiscale spatial and global contextual features for classification. Experimental results on Indian Pines, Houston2013, WHU-Hi-LongKou, and WHU-Hi-HanChuan show that SSH-Net achieves consistently competitive or superior OA, AA, and κ compared with representative baseline methods. Future work will focus on more rigorous analysis of the learned update dynamics, stronger robustness evaluation under broader sensing conditions, and more adaptive propagation scheduling for larger scale and more label-scarce hyperspectral scenarios.

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